Raw data (optional)

HMI / SHARP file names
GOES flare catalog
GOES 1-min X-ray flux

Or your own CSVs

image index: timestamp,
[active_region], [image_path]
flare catalog: start, peak,
end, class, active_region

Input Layer

build_image_index_from_filenames → Image index

adapt_goes_catalog → Flare catalog

add_relative_flux → + rxfi column

build(image index, flare catalog)

Configuration Layer

Dataset class

FullDiskSingleImage
FullDiskSequence
ActiveRegionSingleImage
ActiveRegionSequence

Settings

prediction_window, event_time
cadence, start
observation_window, sliding_window
missing_active_region

Label (strategy)

reducer + labeler, or
BinaryThresholdStrategy,
MaxFlareStrategy, your own

the same label rule for every sample

Labeling Layer

Each image or sequence

t = the image's time,
or the last image's time

EventMatcher

flares starting in [t, t + w)
(only the image's region
for active regions)

Reducer

Max / Min / MeanFlux
ClassCount
CumulativeClassIndex
Max / SumRelativeFlux

Labeler

BinaryThreshold
NumericThreshold
Regression, Multiclass
ClassCount / Presence

one row per image or sequence

Output Layer

Labeled table (pandas DataFrame), one row per image or sequence

single images: timestamp, [active_region], label sequences: sequence_start, sequence_end, timestamps, n_images, [active_region], label
optional: flares (include_flares=True), image_path and other index columns (include_metadata=True)
verbose=True prints a summary; sequences can also return a SequenceBuildReport